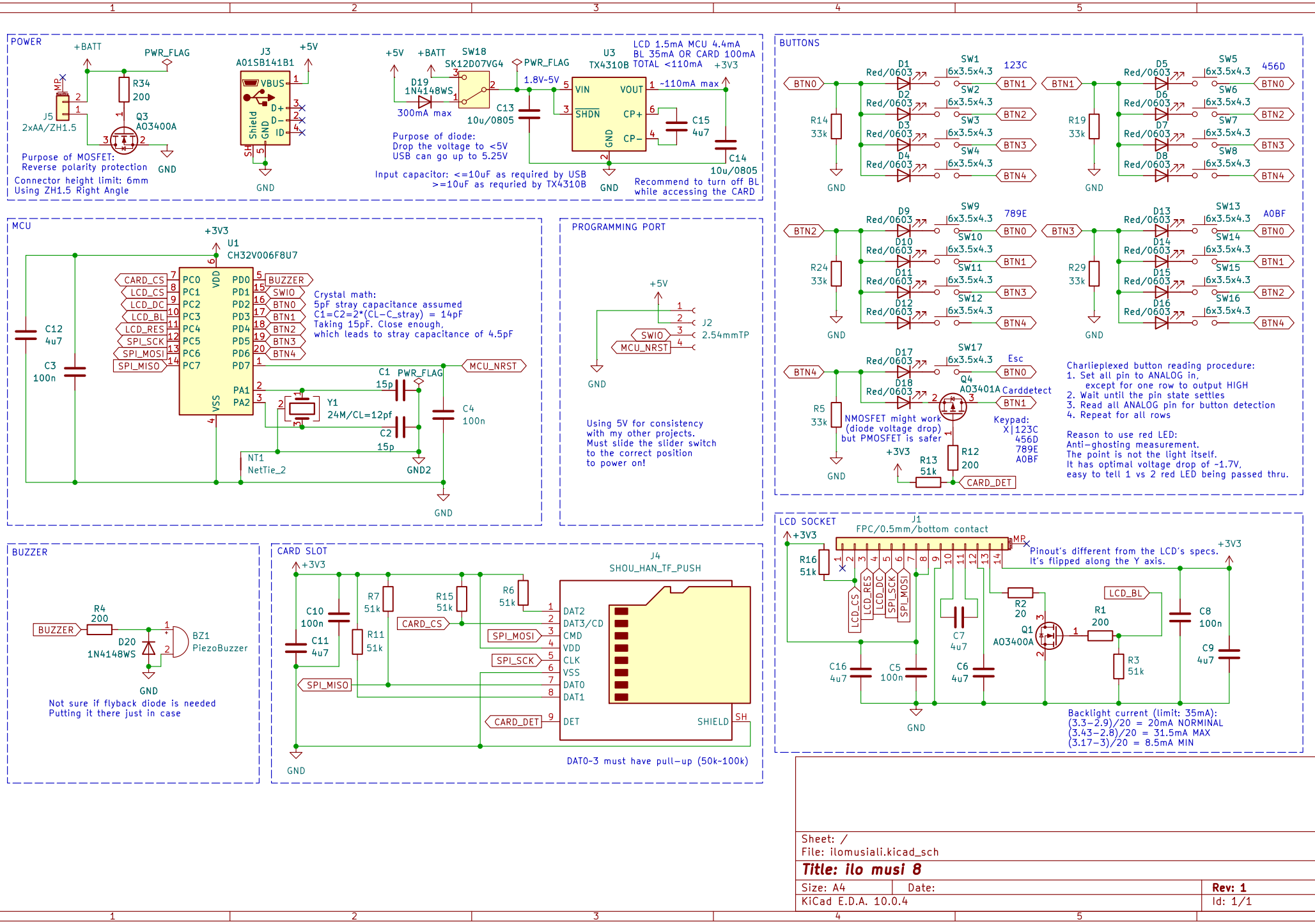

POWER
+BATT
PWR_FLAG
R34
200
Q3
AO3400A
J5
2xAA/ZH1.5
Purpose of MOSFET:
Reverse polarity protection
GND
Connector height limit: 6mm
Using ZH1.5 Right Angle
J3
A01SB141B1
+5V
VBUS
D+
D-
ID
Shield
GND
SH
GND
+5V
+BATT
SW18
SK12D07VG4
PWR_FLAG
D19
1N4148WS
300mA max
C13
10u/0805
1.8V~5V
U3
TX4310B
VIN
VOUT
SHDN
CP+
CP-
GND
~110mA max
C15
4u7
C14
10u/0805
LCD 1.5mA MCU 4.4mA
BL 35mA OR CARD 100mA
TOTAL <110mA +3V3
Purpose of diode:
Drop the voltage to <5V
USB can go up to 5.25V
Input capacitor: <=10uF as required by USB
>=10uF as requried by TX4310B
Recommend to turn off BL
while accessing the CARD
MCU
+3V3
U1
CH32V006F8U7
CARD_CS
LCD_CS
LCD_DC
LCD_BL
LCD_RES
SPI_SCK
SPI_MOSI
SPI_MISO
PC0
PC1
PC2
PC3
PC4
PC5
PC6
PC7
VDD
VSS
PD0
PD1
PD2
PD3
PD4
PD5
PD6
PD7
PA1
PA2
BUZZER
SWIO
BTN0
BTN1
BTN2
BTN3
BTN4
MCU_NRST
C12
4u7
C3
100n
C1
15p
C2
15p
Y1
24M/CL=12pf
C4
100n
NT1
NetTie_2
GND2
GND
Crystal math:
5pF stray capacitance assumed
C1=C2=2*(CL-C_stray) = 14pF
Taking 15pF. Close enough,
which leads to stray capacitance of 4.5pF
PROGRAMMING PORT
+5V
SWIO
MCU_NRST
J2
2.54mmTP
GND
Using 5V for consistency
with my other projects.
Must slide the slider switch
to the correct position
to power on!
BUTTONS
BTN0
BTN1
BTN2
BTN3
BTN4
R14
33k
R19
33k
R24
33k
R29
33k
R5
33k
D1
D2
D3
D4
D5
D6
D7
D8
D9
D10
D11
D12
D13
D14
D15
D16
D17
D18
Red/0603
SW1
SW2
SW3
SW4
SW5
SW6
SW7
SW8
SW9
SW10
SW11
SW12
SW13
SW14
SW15
SW16
SW17
6x3.5x4.3
123C
456D
789E
A0BF
Esc
Carddetect
Q4
AO3401A
NMOSFET might work
(diode voltage drop)
but PMOSFET is safer
Keypad:
X|123C
456D
789E
A0BF
+3V3
R12
200
R13
51k
CARD_DET
GND
Charlieplexed button reading procedure:
1. Set all pin to ANALOG in,
except for one row to output HIGH
2. Wait until the pin state settles
3. Read all ANALOG pin for button detection
4. Repeat for all rows
Reason to use red LED:
Anti-ghosting measurement.
The point is not the light itself.
It has optimal voltage drop of ~1.7V,
easy to tell 1 vs 2 red LED being passed thru.
BUZZER
R4
200
BUZZER
D20
1N4148WS
BZ1
PiezoBuzzer
GND
Not sure if flyback diode is needed
Putting it there just in case
CARD SLOT
+3V3
J4
SHOU_HAN_TF_PUSH
C10
100n
C11
4u7
R7
51k
R11
51k
R15
51k
R6
51k
CARD_CS
SPI_MOSI
SPI_SCK
SPI_MISO
CARD_DET
DAT2
DAT3/CD
CMD
VDD
CLK
VSS
DAT0
DAT1
DET
SHIELD
SH
GND
DAT0~3 must have pull-up (50k~100k)
LCD SOCKET
J1
FPC/0.5mm/bottom contact
+3V3
R16
51k
LCD_CS
LCD_RES
LCD_DC
SPI_SCK
SPI_MOSI
C7
4u7
R2
20
Q1
AO3400A
R1
200
LCD_BL
R3
51k
C8
100n
C9
4u7
C16
4u7
C5
100n
C6
4u7
GND
Pinout's different from the LCD's specs.
It's flipped along the Y axis.
Backlight current (limit: 35mA):
(3.3-2.9)/20 = 20mA NORMINAL
(3.43-2.8)/20 = 31.5mA MAX
(3.17-3)/20 = 8.5mA MIN
Sheet: /
File: ilomusiali.kicad_sch
Title: ilo musi 8
Size: A4
Date:
Rev: 1
KiCad E.D.A. 10.0.4
Id: 1/1